

Verified model update

Updated 2 October 2026. This addendum supersedes the initial guide where it labels all model integrations untested.

All 20 unit and HTTP integration tests passed. They cover actual screenshot OCR, PDF ingestion, request validation, ticket approval, and a public demo mode that avoids saving visitors' input on the server.

Three live AI component tests passed: MiniLM selected the correct login source for a paraphrased question; Whisper tiny transcribed the public JFK audio fixture; Qwen2.5-0.5B-Instruct generated a payment support draft. Detailed machine-readable results are in reports/model_validation.json.

The audio failure was corrected by pinning PyAV to 15.1.0. The CPU generation backend now labels the policy supplied to the model when the model omits a citation. This label identifies the evidence input; it does not prove every claim is supported. The agent must review the response.

These are small component smoke tests, not a broad accuracy, security or customer-satisfaction benchmark. The optional Ollama adapter remains untested. The tested generation backend is Transformers.

GitHub:

<https://github.com/harshakavali81-collab/A-Multimodal-AI-Customer-Support-Copilot> .

The regular Actions workflow runs tests, baseline evaluation, a Docker build and container health check. The separate Live model tests workflow can repeat model inference.

Run the full AI mode and deploy

Windows: install Python 3.12, extract the project and double-click start_models.bat. The script creates a virtual environment, installs CPU dependencies, downloads the three models and launches the application. Install Tesseract separately for screenshot OCR.

Linux: from the project folder, run `bash start_models.sh` with Python 3.11 or 3.12 installed. Initial setup needs internet access and several GB of RAM and disk space. Model weights are cached separately and are not included in the ZIP.

Open `http://127.0.0.1:8000` and select Use AI drafting. Upload a short audio recording to exercise transcription, or ask a paraphrased policy question to exercise neural retrieval. Read the supporting policy before approving any response.

Public demo mode uses `PUBLIC_DEMO=1`. The server saves no customer requests or review history. The page keeps only its own temporary review history, which disappears when it closes. Local mode retains SQLite history.

`render.yaml` prepares a lightweight text/PDF/OCR deployment. `Dockerfile.models` prepares the larger model-enabled container and downloads weights during its build. A sufficiently sized hosting plan is needed for the model image; the lightweight free service is not configured to run all three models.

A connected hosting account is still required. Render was not connected in this session, so no live public application URL is claimed. The source repository is public and the deployment configuration is included. See `docs/DEPLOYMENT.md` for exact instructions and current limitations.